

NutriTrack App Software requirements specifications

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document describes the functional and non-functional requirements for NutriTrack, a mobile application designed to assist healthcare workers in screening and managing child malnutrition cases.

1.2 Scope

NutriTrack is a mobile application that uses Mid-Upper Arm Circumference (MUAC) and other anthropometric indicators to help identify children at risk of acute malnutrition, monitor growth trends, and facilitate timely referrals to nearby health centers. The system integrates with FHIR (Fast Healthcare Interoperability Resources) APIs to store and retrieve patient data in a standardized format, enabling data exchange with national health information systems.

1.3 Definitions, Acronyms, and Abbreviations

- **MUAC:** Mid-Upper Arm Circumference (key indicator for malnutrition).
- **FHIR:** Fast Healthcare Interoperability Resources (HL7 standard for health data exchange).
- **WHO:** World Health Organization (provides malnutrition classification standards, e.g., MUAC <11.5cm = severe acute malnutrition for children 6-59 months).
- **HTTPS:** Hypertext Transfer Protocol Secure.
- **API:** Application Programming Interface.
- **SDK:** Software Development Kit (e.g., Android SDK).
- **DFD:** Data Flow Diagram.
- **UML:** Unified Modeling Language.
- **HL7:** Health Level Seven International (FHIR standard body).

2. Functional Requirements

FR1: Patient Management

The system shall allow users to register, update, and view patient profiles using the FHIR Patient resource.

FR2: Observation Entry

The system shall enable recording of MUAC and weight-for-height measurements using the FHIR Observation resource.

FR3: Condition Detection

The system shall automatically determine malnutrition severity using the FHIR Condition resource based on WHO standards.

FR4: Referral System

The system shall create ReferralRequest or ServiceRequest resources for critical malnutrition cases.

FR5: Nutrition Orders

The system shall generate dietary or supplement recommendations using the FHIR NutritionOrder resource.

FR6: Synchronization

The system shall sync offline data to the FHIR server when network connectivity is restored.

FR7: Reporting

The system shall generate malnutrition and referral summary reports.

FR8: User Management

The system shall implement secure authentication and role-based access control for Admin and Health Worker roles.

FR9: Growth Tracking

The system shall visualize growth and nutritional progress over time.

FR10: Offline Functionality

The system shall cache data locally for areas with limited internet connectivity.

3. Non-Functional Requirements

3.1 Performance Requirements

- Patient data must load within 2 seconds
- The system shall support 1,000+ patient records per community health center

3.2 Security Requirements

- All API communications shall be secured using HTTPS
- The system shall implement role-based access control

3.3 Usability Requirements

- The system shall use Material Design principles for user interface
- MUAC measurements shall use color-coded classification (Green, Yellow, Red zones)

3.4 Reliability Requirements

- The system shall provide offline functionality with automatic synchronization when connectivity is restored

3.5 Compatibility Requirements

- The system shall be compatible with Android devices running SDK 24 or higher
- The system shall comply with HL7 FHIR API standards

3.6 Scalability Requirements

- The system shall support 1,000+ patient records per community health center

4. External Interface Requirements

4.1 Software Interfaces

The system shall integrate with the following FHIR resources:

FHIR Resource	Purpose
Patient	Store and manage child demographic and health details
Observation	Record anthropometric data (MUAC, weight-for-height)
Condition	Determine and record malnutrition diagnosis

ReferralRequest/ServiceRequest Create and manage referrals for children needing advanced care

NutritionOrder Recommend dietary interventions or supplements

NutriTrack Use Case Diagram

Description

The NutriTrack system is a mobile application designed to assist healthcare workers in screening and managing child malnutrition cases. The use case diagram illustrates the interactions between the system and its primary actors, showing the core functionalities of the application.

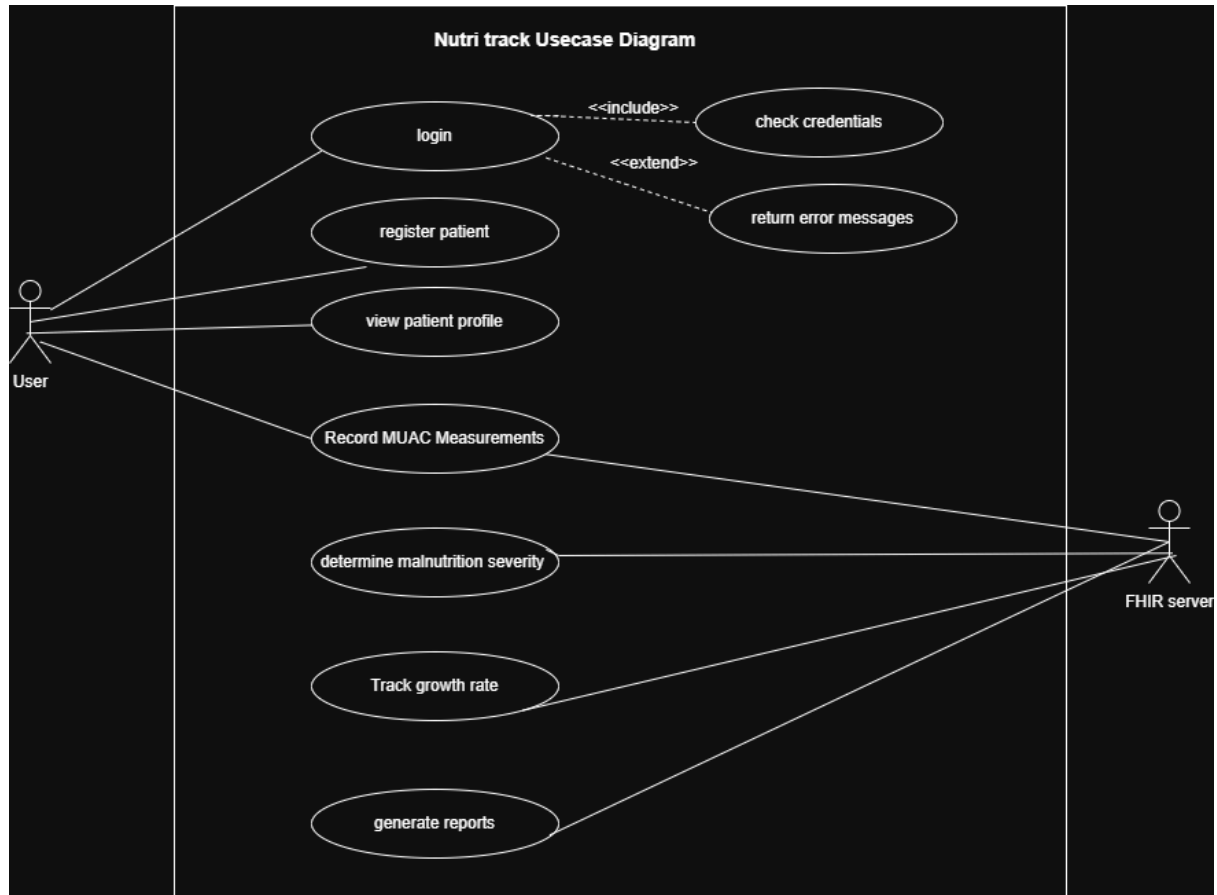
Actors

1. **Nurse** – A healthcare worker responsible for registering patients, recording MUAC measurements, tracking growth, and syncing data to the FHIR server.
2. **Doctor** – Reviews diagnosis, creates referrals, prescribes nutrition orders, and analyzes reports. Positioned below the Nurse in the diagram to indicate hierarchical workflow.
3. **Patient** – Can view their own health records through the system.
4. **FHIR Server** – External system used for storing and retrieving standardized health data.

Use Cases

1. **Register Patient** – Captures and manages child demographic and health details.
2. **Record MUAC Measurement** – Allows nurses to record mid-upper arm circumference values for malnutrition screening.
3. **Track Growth** – Monitors child growth trends over time.
4. **Sync Data (FHIR)** – Synchronizes offline or collected data with the external FHIR server.
5. **Review Diagnosis** – Doctors review automatic or manual diagnosis results.
6. **Create Referral** – Generates referral requests for children requiring further care.
7. **Prescribe Nutrition Order** – Recommends diet plans or supplements based on diagnostic outcomes.
8. **Analyze Reports** – Provides doctors and supervisors with malnutrition trends and statistics.
9. **View Own Health Records** – Allows patients to access their personal health information.

Diagram:



Nutri track DataFlow Diagram

System Purpose & High-Level Flow

The primary purpose of NutriTrack is to support Health Workers in identifying and managing cases of child malnutrition efficiently. The process flow, as shown in the DFD, can be summarized as follows:

1. A Child/Caregiver provides demographic information during Patient Registration (1.0)
2. A Health Worker performs a MUAC Measurement (2.0) and inputs the data
3. The system automatically performs a Diagnosis Classification (3.0) based on the measurement

4. Based on the diagnosis:

- Severe cases are sent to Referral Management (5.0) to be directed to a Health Center
- Cases requiring nutritional support trigger Nutrition Orders (6.0)

5. All MUAC data is used for Growth Tracking (4.0) over time, which feeds into Reporting & Analytics (8.0) for Supervisors/Admins

6. All data is stored locally (D6: Local Cache) and synchronized with a central FHIR Server (7.0) to ensure data persistence and accessibility

Core Processes

The DFD outlines core processes:

- 9.0 User Authentication: Secures system access for Health Workers
- 1.0 Patient Registration: Captures and stores basic patient details
- 2.0 MUAC Measurement: Records the Mid-Upper Arm Circumference data
- 3.0 Diagnosis Classification: Automatically classifies the nutritional status based on MUAC
- 4.0 Growth Tracking: Monitors and visualizes a child's growth trajectory over time
- 5.0 Referral Management: Creates and manages referrals to higher-level health facilities
- 6.0 Nutrition Orders: Manages the prescription and tracking of nutritional supplements
- 8.0 Reporting Analytics: Generates reports and dashboards for oversight
- 7.0 Data Sync: Synchronizes data between the local device and a central FHIR server

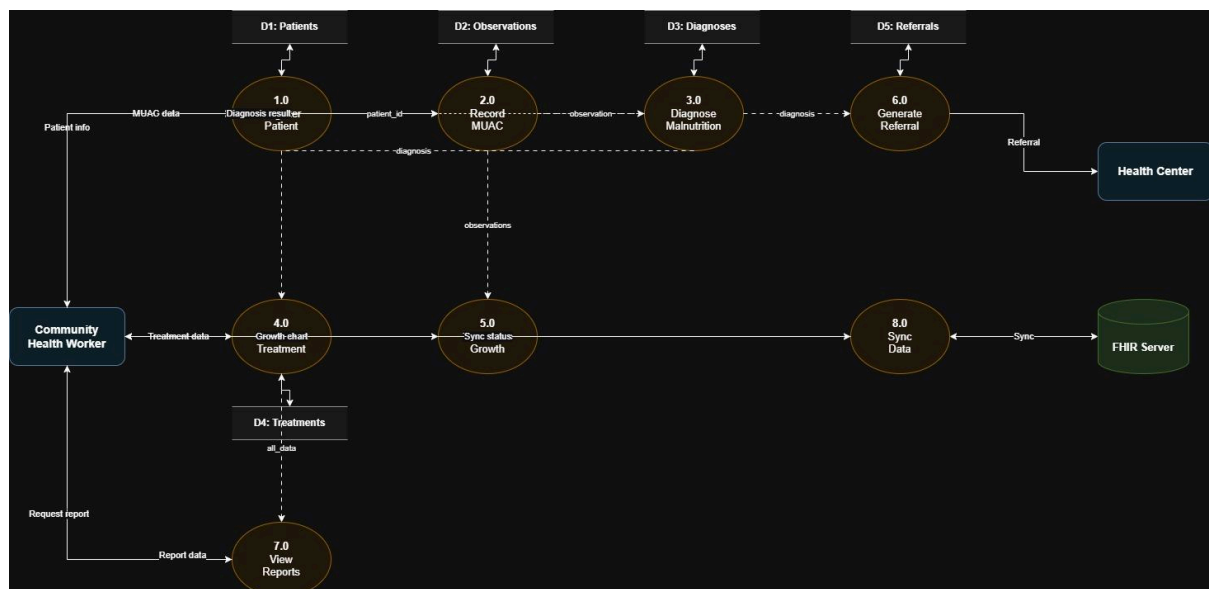
Data Stores

The system utilizes several data stores to persist information:

- D1: Patient Records: Stores demographic and personal information
- D2: Observations: Holds all MUAC measurement data
- D3: Conditions: Stores the diagnosis and classification results
- D4: Referrals: Maintains records of all referral cases
- D5: Nutrition Orders: Keeps a log of all nutritional orders prescribed
- D6: Local Cache: Temporarily holds all data for offline functionality before syncing
- D7: Users & Roles: Manages system user accounts and permissions

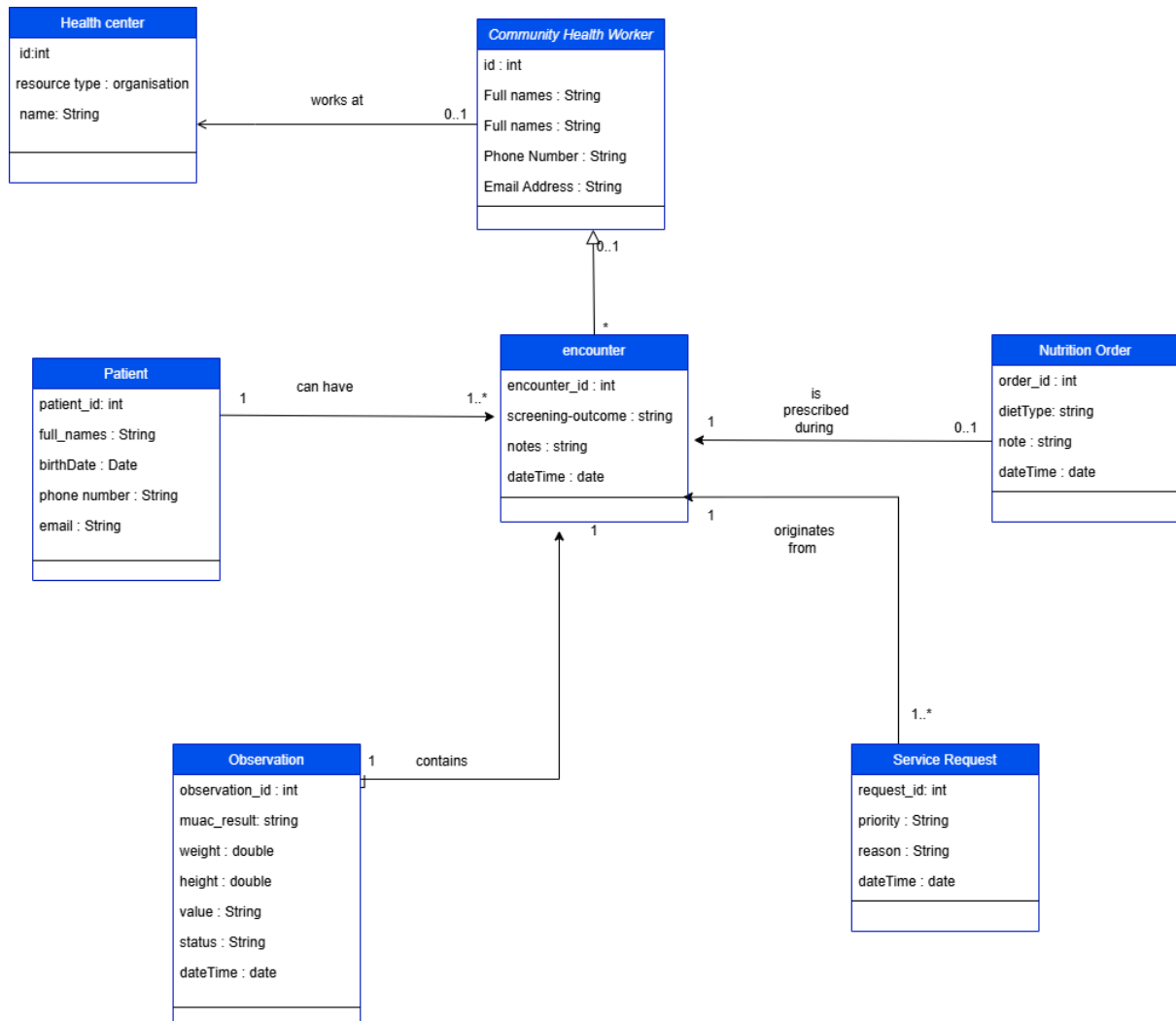
The NutriTrack DFD illustrates a robust and integrated system that digitizes the frontline process of combating child malnutrition. It ensures that data flows seamlessly from initial patient contact through to diagnosis, treatment, referral, and program analysis, all while supporting offline use and interoperability through the FHIR standard.

Diagram:



NutriTrack UML Class Diagram:

Image:



The above picture provides a brief explanation of the **NutriTrack UML Class Diagram**.

The diagram illustrates the interaction between FHIR-based resources and locally defined application classes. It shows how patient data, observations, nutrition orders, service requests, and conditions are structured using FHIR R4 models. These classes include key attributes such as identifiers, status, coding, and references.

The local application classes (User, HealthCenter, Report, LocalCache, FhirSyncService) demonstrate how the app manages offline data, synchronization, user authentication, reporting, and referral destinations.

The main purpose of the diagram is to:

- Represent the structure of FHIR resources used in the system (Patient, Observation, NutritionOrder, Condition, ServiceRequest).
- Show relationships between resources such as patient → observation, patient → nutrition order, and condition → clinical status.
- Illustrate how the local components interact with FHIR data, enabling offline-first functionality, data caching, and PDF reporting.
- Provide a clear overview of how information flows between classes for nutrition tracking and health service referrals.

Color meanings shown in the diagram:

- Blue classes: FHIR R4 resources (current)
- Green classes: Local system components (future or application-specific)

This class diagram serves as a blueprint for both backend data modeling and application logic implementation.

Link to the Prototype:

<https://www.figma.com/make/ISX7LCzX6VQBO0GTVV55J6/NutriTrack-UI-Overview?node-id=0-1&p=f&t=phdEajf8gzoW7j0n-0&fullscreen=1>

Meeting Minutes

Date: Monday, 17 November 2025

Time: 10PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: valentin

1. Agenda

- Review of prototype progress
- Discussion on diagram revisions
- Planning for implementation phase

2. Discussion Summary

- The current **prototype** was reviewed and deemed in need of significant changes.
- It was agreed that the **updated prototype** must be completed **by the end of 18 November 2025**.
- The following diagrams were also reviewed and require updates:
 - **Use Case Diagram**
 - **Data Flow Diagram (DFD)**
 - **Class Diagram**
- All diagram revisions are to be finalized **by the end of tuesday 18 november 2025**.
- The meeting will take place daily at 10 PM.

3. Next Steps

- **Implementation(coding) phase** will begin **on Wednesday (19 November 2025)**.
- The team will start by developing **layouts and activities** based on the finalized prototype.

Meeting Minutes

Date: Tuesday, 18 November 2025

Time:10PM

Location:Online Google Meet

Attendees: 4 participants

Recorder: valentin

1. Agenda

- Review of prototype progress
- Discussion on the updated diagrams

2.Discussion summary

- The team agreed that the prototype shall be completed no later than Wednesday(**19 November 2025**) at 12:00 AM.
- Upon completion of the prototype, work will commence on the **layouts and activities** for the mobile application

- **Task assignments** will be distributed to all team members **after the prototype is finalized.**

On **Wednesday and Thursday**, the meeting was conducted via Whatsapp Call just to talk about how each member is doing with different tasks assigned.

Meeting Minutes

Date: Saturday, 22 November 2025

Time: 9PM

Location: Online Google Meet

Attendees: 6 Participants

Recorder: Valentin

1. Agenda

- Review of User Interface(UI) progress
- Discussion on Database design

2. Discussion summary.

- The team agreed that the User Interface shall be completed no later than Sunday(**23 November 2025**) at 12:00 PM.
- Upon completion of the User Interface, we will make a research on the APIs for the mobile application
- And look on the Database design

Meeting Minutes

Date: Monday, 1st December 2025

Time: 10PM

Location: Online Google Meet

Attendees: 4 Participants

Recorder: Kabano

1. Agenda

- Review on the Class Diagram
- Discussion on the Backend

Discussion Summary

- The team shall work on the class diagram
- Upon completion of the Class Diagram, we will work on the backend

Meeting Minutes: Prototype to the user interface Implementation in android studio

Date:23/11/2025

Project: Nutri Track

Attendees:7

Recorder: valentin

Discussion Points & Progress

- **Prototype Features Implemented & Working:**

- User Login – authentication flow tested and functional.
- Signup Page – new user registration working as expected.
- Dashboard – accessible after login, displaying core app features.
- Add Patient Module – patient data entry and storage functional.
- User Details Page – profile information view and update working.

- **Remaining Work:**
- User Settings Integration.

Decisions Made

- Confirmed that all core modules (login, signup, dashboard, add patient, user details) are stable.
- Agreed that merging User Settings is the final step before declaring prototype completion.
- Assigned responsibility for reviewing and merging the local branch into **dev**.

Next Steps

- **Developer Action:**
 - Conduct code review of the local branch containing user settings.
 - Resolve any merge conflicts with the remote **dev** branch.
 - Push changes and confirm successful integration.
- **Team Action:**
- Perform final round of testing once user settings are merged.
- Prepare documentation for prototype handoff.

Summary

The prototype is nearly complete with all major features functioning. The only pending task is merging the User Settings module from the local branch into the remote **dev** branch. Once merged and tested, the prototype will be ready for the next phase(database).

Meeting Minutes

Date: Tuesday, 2 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

- Finalization of class diagram for presentation
- Separation of UI tasks for GitLab push
- Review of pending features

2. Discussion Summary

The class diagram was finalized and prepared for presentation.

Settings and other pending features were successfully pushed to GitLab.

3. Next Steps

- On 3 December, discuss additional UI features to be added.

Meeting Minutes

Date: Wednesday, 3 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

- Identification of new UI features
- Planning immediate development tasks

2. Discussion Summary

The team discussed and agreed to add the following modules:

- Encounter
- Observation
- Nutrition Order
- Service Requests

Development work on these modules began immediately.

3. Next Steps

- On 4 December, verify integration of newly added features.

Meeting Minutes

Date: Thursday, 4 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

- Verification of newly added features
- Exploration of further enhancements

2. Discussion Summary

All discussed features (Encounter, Observation, Nutrition Order, Service Requests) were successfully integrated.

The team brainstormed potential improvements and additional functionalities for these modules.

3. Next Steps

- On 6 December, begin database creation to support patient and encounter data storage.

Meeting Minutes

Date: Saturday, 6 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

- Database design and implementation

2. Discussion Summary

A SQLite database was created to store:

- Patient records
- Encounters
- Other related entities for efficient data retrieval.

3. Next Steps

- On 7 December, discuss schema design to align system with real-world hospital structures.

Meeting Minutes

Date: Sunday, 7 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

- Structuring system with hospital schema

2. Discussion Summary

The team discussed integration of a hospital schema to ensure:

- System reflects real-world healthcare workflows.
 - Health workers operate under specific hospital contexts.
- This schema will improve scalability and organizational clarity.

3. Next Steps

- On 8 December, draft schema implementation plan and assign responsibilities for integration into the system.

Meeting Minutes

Date: Monday, 8 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 7 participants

Recorder: Valentin

1. Agenda

FHIR API generation and initial testing

2. Discussion Summary

- Generated FHIR APIs for healthcare data exchange.
- Tested APIs using Postman to confirm data could be saved on the FHIR server.

3. Next Steps

- Document API endpoints for developer reference.
- Implement authentication and authorization for secure access.
- Prepare integration of Patient and Encounter modules.

Meeting Minutes

Date: Wednesday, 10 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

Integration of Patient and Encounter features

2. Discussion Summary

- Integrated Patient and Encounter modules with FHIR APIs.
- Tested functionality to ensure smooth interaction.

3. Next Steps

- Expand integration to Observation, Nutrition Order, and Service Request modules.
- Develop automated test scripts for regression testing.
- Optimize API performance for large datasets.

Meeting Minutes

Date: Thursday, 11 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 5 participants

Recorder: Valentin

1. Agenda

Follow-up on Patient/Encounter testing

2. Discussion Summary

- Meeting held but no major technical milestones documented.

3. Next Steps

- Address pending bug fixes from Patient/Encounter testing.
- Groom backlog for upcoming features.

Meeting Minutes

Date: Saturday, 13 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 8 participants

Recorder: Valentin

1. Agenda

Expansion of NutriTrack APIs

2. Discussion Summary

- Created and integrated APIs for Observation, Nutrition Order, and Service Request.
- Connected these APIs to the NutriTrack application.

3. Next Steps

- Validate data consistency across modules.
- Add UI components to display new clinical data.
- Conduct end-to-end testing with sample patient records.

Meeting Minutes

Date: Sunday, 14 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 6 participants

Recorder: Valentin

1. Agenda

Multilingual support implementation

2. Discussion Summary

- Added language filter enabling French, English, and Kinyarwanda support.

3. Next Steps

- Expand language support to additional regional languages.
- Conduct usability testing with multilingual users.
- Implement dynamic translation for clinical terms.

Meeting Minutes

Date: Monday, 15 December 2025

Time: 10 PM

Location: Online Google Meet

Attendees: 7 participants

Recorder: Valentin

1. Agenda

Server synchronization and chart updates

2. Discussion Summary

- Implemented sync-to-server functionality to upload offline data once connectivity is restored.
- Updated charts to reflect real system data.

3. Next Steps

- Monitor sync performance under varying network conditions.
- Add error-handling and retry mechanisms for failed sync attempts.
- Enhance chart customization options for different user roles.